

General idea:

Building an product that transforms NL prompts to Playwright test using MCP server.

Steps:

- Research abt the available tools and used techs
- First steps in tools we are using
- Creating demo
- Preping materials to show advantages of out tool

Research results:

POM - Page as Object Model

It involves creating class for each page of application, containing the elements of page and ways to interact with them. Widely considered a “best practice” for web-pages testing using tools like Playwright or Selenium, making code clean and easy to maintain.

In context of our product using POM, should greatly reduce AI models hallucinations and maintain reliable test generation through the whole testing process. Another advantage is possible token conservation.

MCP clients comparision

Client with consider are:

- VS Code
- Cursor
- Windsurf
- Claude Desktop
- Goose

Tool name	Setup ease	Browser integration	Main advantage
VS Code	High (extension)	Integrated	Lately introduced integrated browser
Cursor	Medium (JSON)	External	Its a tool optimized for working with your existing code
Windsurf	Highest (one-click)	External	Extremely easy integration with Playwright
Claude Desktop	Low (manual config)	External	Its better choice if you want non-IDE tool
Goose	Medium (extensions)	External	Optimized for Playwright CLI, token efficient, possible automation of flaky tests*

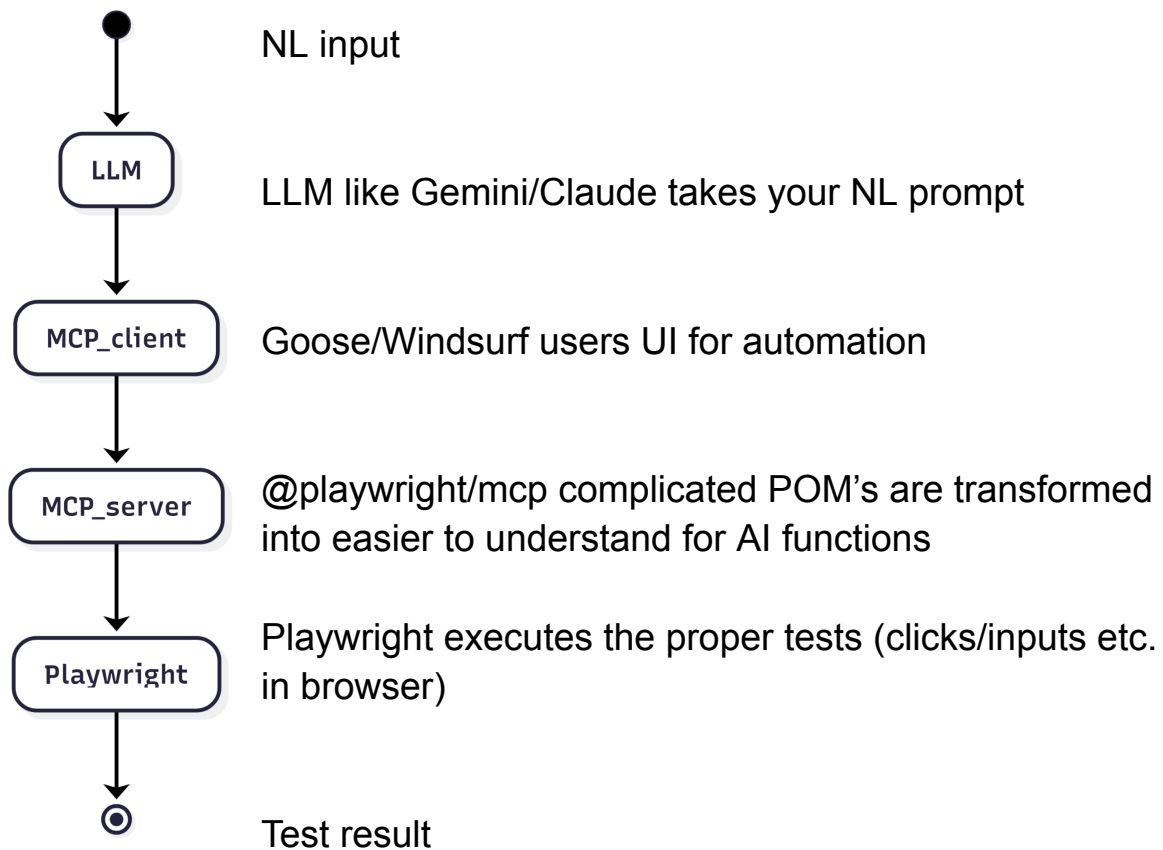
flaky test - unstable, sometimes passed, sometimes not

For early testing of a solution **Windsurf** is a good choice because of easy setup, but our target should be **Goose** with the test automation and Playwright CLI usage to make the tool token-efficient.

Another considerable choices are

- **VS Code** because of popularity and integrated browser.
- **Cursor** in environment where a lot of similar test where written to keep testers style of coding

Integration diagram for the tool



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AI vs Human cost comparison

If we consider a wage of a mid/senior QA tester at 100PLN/hr, and estimated time to write test manually at 4 hrs, cost of test will be 400PLN.

AI on the other hand will take significantly less time (around 2-5 min) and cost with an assumption of usage of 5-15 k tokens will be 0.05\$ - 0.25\$.

If we consider that around 10% of saved time is used on the AI generated tests auditing, we can save abt 90% of time and money spent on test.

Competition analysis

Tool	Privacy/security	Cost efficiency	Addition
PlaywrightCLI	High	Low token usage	Large scale/long sessions
Browser use	If paired with Ollama, highest bcs of local model	Free in terms of tokens, bcs of local AI	Good in high privacy environments
Stagehand	Medium (session caching, cloud first)	Medium	The most reliable

We didn't consider Playwright MCP, CLI version is much more token efficient (around 4-times) and gives the same testing abilities.